

R: The language for data science

Data analysis and insights are urgently needed because of the volume of information available. Raw data is transformed into finished data products with the help of a number of technologies. When it comes to research, processing, transformation, and visualisation of data, there are few better tools than R.

The key features in R for data science are:

- Data pre-processing
- Social media fetching and analysis
- Assorted operations on data frames
- Extract, transform, load (ETL)
- Connection to a variety of databases, including SQL and spreadsheets
- Interaction with NoSQL databases
- Training and prediction with models
- Machine learning models
- Clustering
- Fourier transform
- Web scraping

R is a robust programming language commonly used in statistical computation and data analysis. Efforts to enhance R's user interface have been going on since long. Several data science groups throughout the world have followed the development of R from a simple text editor to the more current interactive R Studio and Jupyter Notebooks.



Figure 1: Key applications and use cases of data science

Only the contributions of R users throughout the world have made this possible. The inclusion of strong packages in R has made it increasingly powerful over time. Numerous open source packages have made it easier and more efficient to work with large data sets and visualise data.

Radiant Data Design Basics Model Multivariate Report

Manage View Visualize Pivot Explore Transform Combine

Data preview

price	carat	clarity	cut	color	depth	table	x	y	z	date
580	0.32	VS1	Ideal	H	61.00	56.00	4.43	4.45	2.71	2012-02-26
650	0.34	SI1	Very Good	G	63.40	57.00	4.45	4.42	2.81	2012-02-26
630	0.30	VS2	Very Good	G	63.10	58.00	4.27	4.23	2.68	2012-02-26
706	0.35	VVS2	Ideal	H	59.20	56.00	4.60	4.65	2.74	2012-02-26
1080	0.40	VS2	Premium	F	62.60	58.00	4.72	4.68	2.94	2012-02-26
3082	0.60	VVS1	Ideal	E	62.50	53.70	5.35	5.43	3.38	2012-02-26
3328	0.88	SI1	Ideal	I	61.70	56.00	6.14	6.18	3.80	2012-02-26
4229	0.93	SI1	Premium	E	61.40	57.00	6.34	6.23	3.86	2012-02-26
1895	0.51	VVS2	Very Good	G	63.40	57.00	5.09	5.06	3.22	2012-02-26
3546	1.01	SI2	Good	E	63.90	58.00	6.31	6.37	4.05	2012-02-26

10 of 3,000 rows shown. See View-tab for details.

Datasets: diamonds

☐ Add/edit data description
☐ Rename data

Display:
☒ preview ☐ str ☐ summary

Load data of type:
 rds | rda | rdata

Browse... No file selected

Save data to type:
 rds

Save

Figure 2: Shiny based app